character, if for no other reason than that the neutron carries no electric charge. Furthermore, electric forces would be much too weak to account for the considerable energies which result from the mass defects.

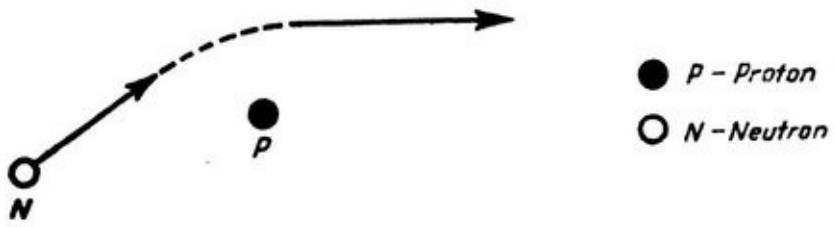

Figure 14.—Deflection of a neutron in the neighbourhood of a proton

We have already pointed out that when a deuteron is formed out of a proton and a neutron, the binding energy is liberated as a photon, with an energy of 2·2 Mev.—in other words, as electromagnetic energy. This means that a process is taking place in which energy is converted from one form to another—the non-electromagnetic energy of the nuclear field into the electromagnetic energy of radiation. Therefore, it follows that in common with all other types of energy, the energy of the nuclear field possesses the capacity of being transformed into other forms of energy.

We can gain some insight into the dependence of nuclear forces on the distance by observing the deflection of flying neutrons when they pass near a proton (Figure 14). Modern physics has access to sources of neutrons. All that is necessary is simply to send the neutrons through a hydrogen-containing substance, for example, a hydrocarbon, such as paraffin, or water, to cause them to be deflected from their straight paths. The magnitude of the deflection of a neutron depends, naturally, on the distance of its path from the proton. This distance is more frequently a greater than a smaller one. Instances of neutrons passing close to a proton are very rare. If the forces decrease relatively slowly with the distance—as, for instance, in the case of electric charges—even though the distance may be considerable, the neutrons will still be deflected a little. It would indeed be observed that a very large number of neutrons are deflected, but the deflection is always very slight. As a matter of fact, large deflections are seldom observed. If, on the other hand, the force diminishes rapidly with the distance, the majority of the neutrons are not deflected at all. In the case of the deflected neutrons—those few which pass sufficiently close to the proton—both small and large deflections may be observed to occur with comparable frequency.

Such experiments have demonstrated that the force between neutron and proton diminishes with the distance more rapidly than is the case with electric forces of attraction and repulsion. The degree of accuracy of the measurements does not as yet permit us to formulate exactly the law of